

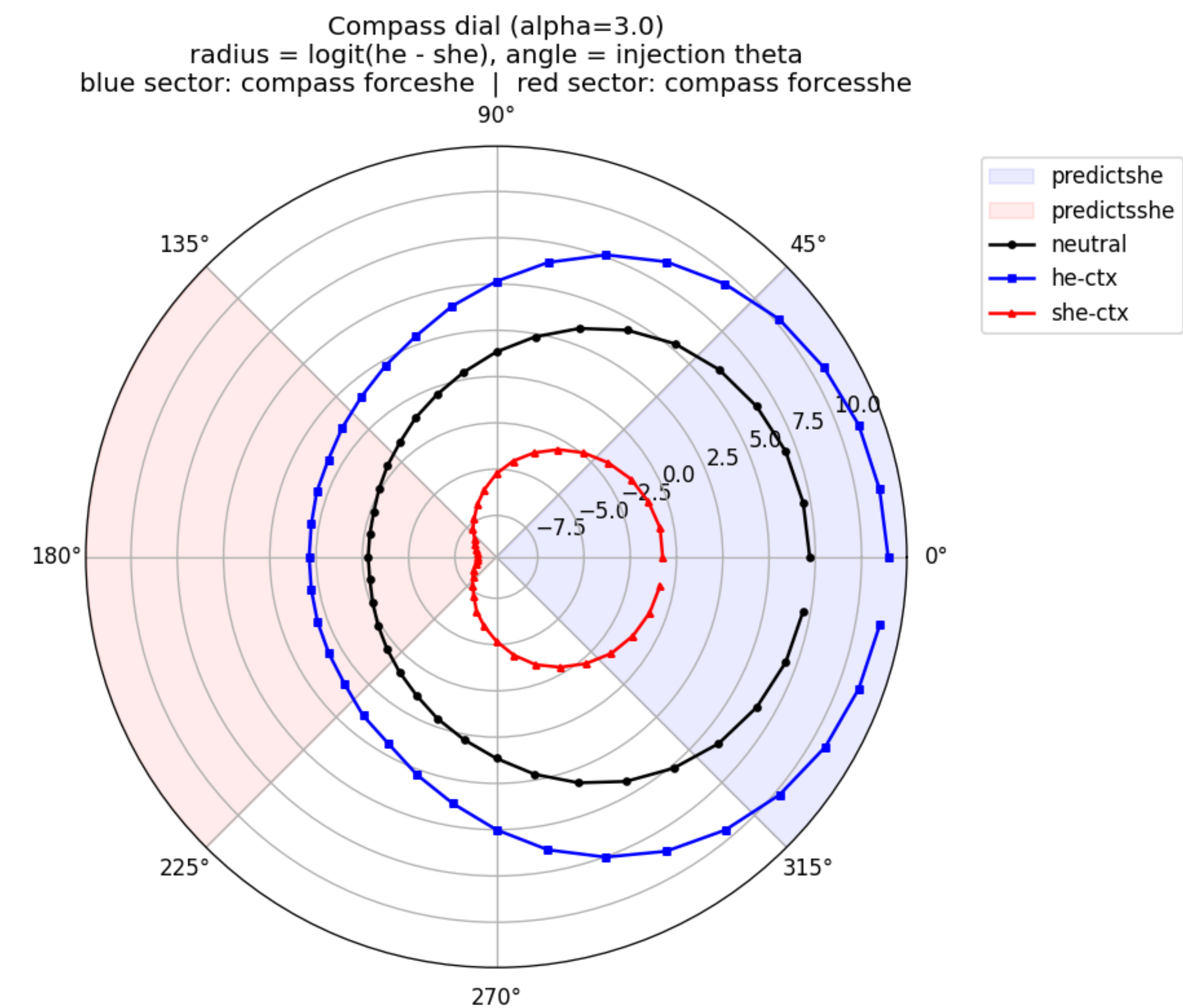
COMPASS: Two-Dimensional Semantic Dials Inside a Single Attention Head

A rank-2 causal editing tool that reads, writes, and routes per-domain bias — GPT-2 · Phi-3-mini · Gemma-2-2B · Llama-3.2-3B.

1. What is a compass?

- For head h at layer ℓ , factor $W_{OV} = W_V W_O = U \Sigma V^\top$.
- A **compass plane** is two right-singular vectors (u_i, u_j) ; inject along it:
 $v(\theta, \alpha) = \alpha \sigma_i \cos \theta u_i + \alpha \sigma_j \sin \theta u_j$.
- If the plane is causal, logit-difference swings sinusoidally in θ and linearly in α .

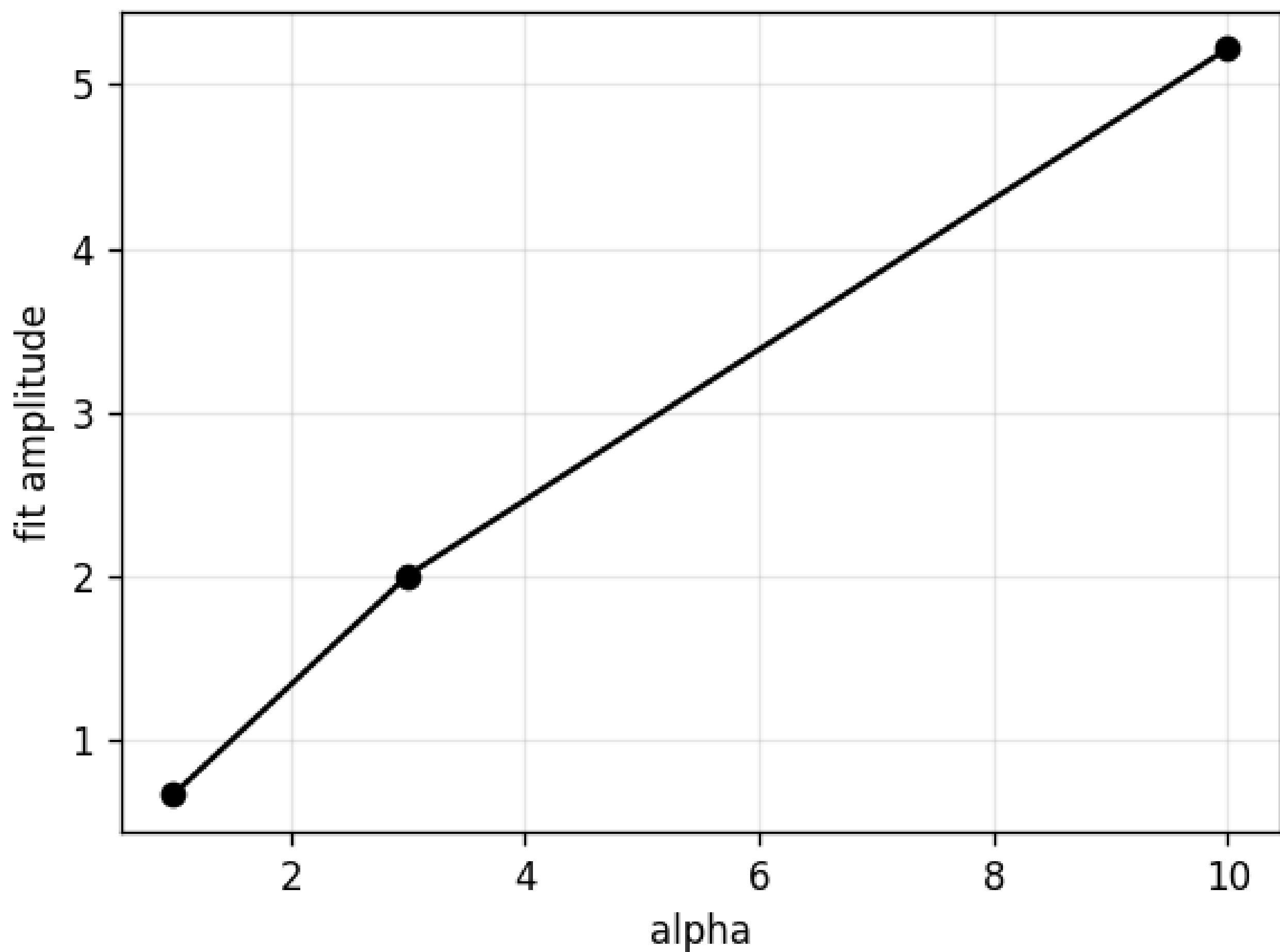
2. Sinusoidal response



Phi-3 Mini L28H1 gender compass. Radius = logit($he - she$), angle = injection θ . Poles at $0^\circ/180^\circ$ decode to $he / his / him$ vs. $she / her / hers$.

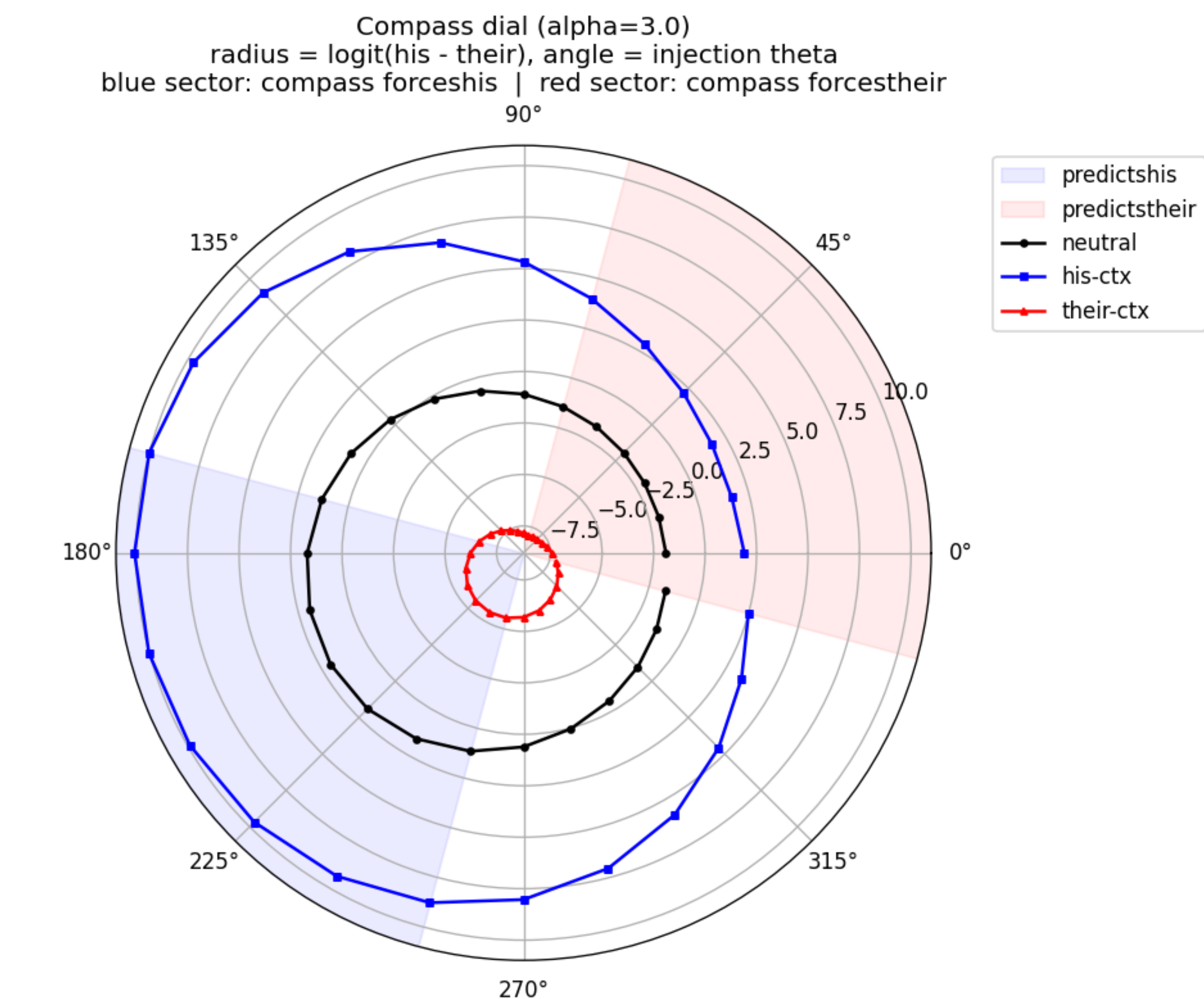
3. Linearity + phase stability

Linearity: amplitude vs alpha



Two-pillar test: amplitude $R^2 \geq 0.993$ vs. α , phase drift $\leq 3^\circ$ across $\alpha \in \{1, 3, 10\}$. Random planes and

4. Not just gender



The same two-vector recipe finds *his / their* (number) and temporal (*was / is*) planes in the same heads. Compasses are not a pronoun-specific artifact.

5. Pipeline



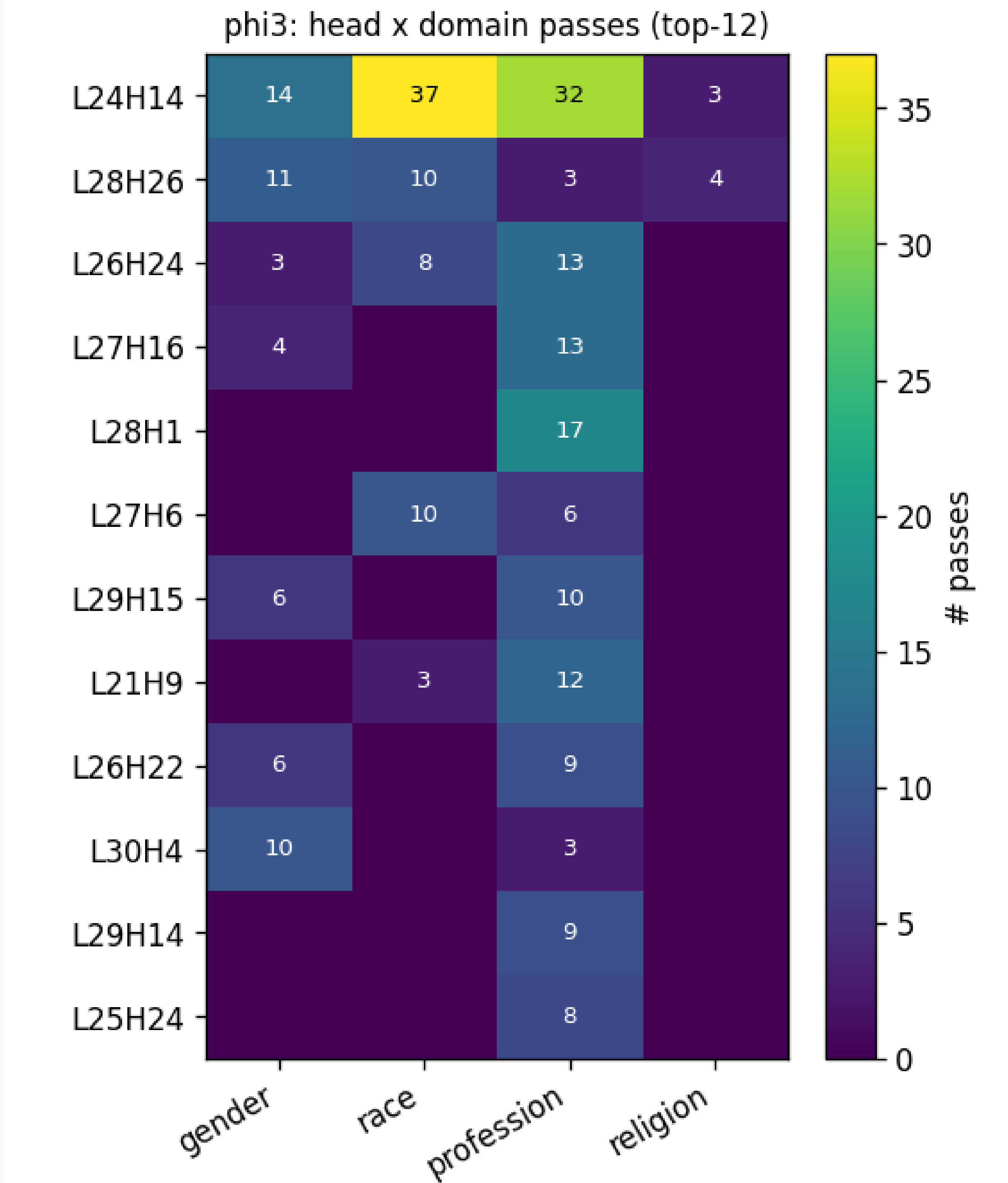
Blind-scan pass rates vs. full-OV null (95% CIs): GPT-2 2.06%, Phi-3 1.18%, Gemma 2.02%, Llama 0.22% — all significantly above the null.

6. Four models, same structure

Model	head	$A(10)$	$\Delta\phi$
GPT-2 Small	L10H9	14.16	0.3°
Phi-3 Mini	L28H1	12.85	1.3°
Gemma-2-2B	L21H4	2.19	1.3°
Llama-3.2-3B	L26H14	3.01	2.4°

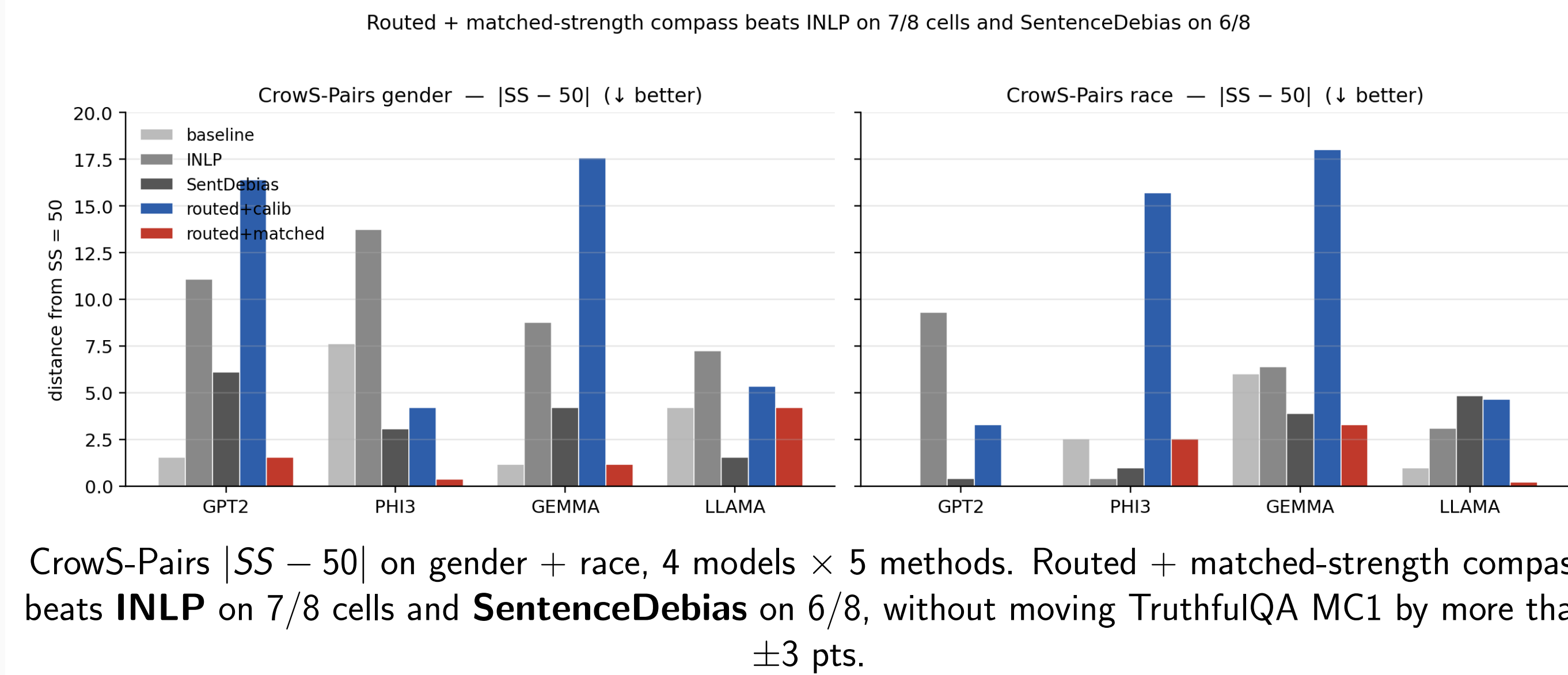
Amplitudes differ by $\sim 5\times$ across families, but the two-pillar recipe finds the same kind of plane in all of them.

7. Head–domain routing



Each passing head specializes for one of {gender, race, profession, religion}. We inject only on rows whose bias type matches, at an SNR-calibrated per-domain $\alpha_d = \tau / \text{SNR}_d$.

8. Debiasing beats INLP / SentenceDebias



CrowS-Pairs |SS = 50| on gender + race, 4 models $\times$ 5 methods. Routed + matched-strength compass beats **INLP** on 7/8 cells and **SentenceDebias** on 6/8, without moving TruthfulQA MC1 by more than ± 3 pts.